

Multiple Choice (10 marks)

1. Gavin has two rocks. Both rocks are made up entirely of the same mineral. What other property of his two rocks is most likely to be the same?
 - (a) size
 - (b) shape
 - (c) color
 - (d) weight
2. Which is an example of a human-made product?
 - (a) cake
 - (b) tomato
 - (c) wheat
 - (d) coal
3. Sulfur (S), oxygen (O₂), water (H₂O), and sodium chloride (NaCl) are all examples of pure substances. Which of the following describes all pure substances?
 - (a) A pure substance consists of only one type of element.
 - (b) A pure substance has a definite chemical composition.
 - (c) A pure substance cannot be broken down into simpler substances.
 - (d) A pure substance is normally found as a solid at room temperature.
4. The speed of a sound wave varies as it travels through different substances. Which factor will most affect the speed of a sound wave?
 - (a) the frequency of the wave
 - (b) the wavelength of the wave
 - (c) the source that created the sound
 - (d) the distance between molecules in the medium
5. A lizard species mostly eats fruit from one particular type of tree. If a virus kills most these trees, which individual lizards will most likely survive?
 - (a) lizards that can climb higher in the fruit trees
 - (b) lizards that have darker coloration
 - (c) lizards that can find other types of food
 - (d) lizards that produce more offspring

True / False (10 marks)

Answer True or False

6. True or False: Heat transferred through the ice cube by convection caused it to melt on the warm sidewalk.
7. True or False: Picking up broken glass pieces by hand is the best way to ensure safety during cleanup.
8. True or False: A proton is not a fundamental component of an atom and can be found outside the nucleus.
9. True or False: Windmills are often built in forests to help protect them from strong winds and storms.
10. True or False: The magnetic force on an object will remain the same regardless of the distance from the magnet.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the impact of deforestation and land burning for agriculture on Earth's atmosphere, including specific consequences for climate and air quality.
12. Discuss how the freezing temperature of water allows it to exist in two different forms on Earth, and analyze the significance of these forms in our environment.

End of Paper